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A New Color-Based Instrument Model For Basic Music Learning

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***Abstract*—The development of applied hardware technology supports efforts on the education and creativity of users and has made microcontrollers a popular device among education practitioners and researchers. The focus of this article is learning for children in the introduction of activities in color-based creativity. A model design of a color-based musical learning instrument device, composed of 2 main elements. The TCS3200 color sensor and arduino microcontroller, used to convert system inputs in the form of unique pieces of color paper arranged on a plastic tape, will be the instrument's modeling keywords. The size of the color paper piece is the same width as the plastic tape, which is 5 centimeters, while the dimensions of length are 5 centimeters to represent the beat of the tone every second. The process begins with color selection of tone characters, color recognition and color classification. Meanwhile, the design of the system algorithm, has the ability to detect colors, determine the tone according to the choice of color categories, and adjust the stability of the speed of the DC motor 60 revolution per minutes. The final test results of the system successfully proved the color detection process of sensor input and tone audio selection according to the classification of specified categories while proving that the system is able to work properly.**

***Index Terms*—Color Sensor, Arduino controller, RGB, Color band-tones**

I. Introduction

The development of the world of technology, in the use of internet of things devices, has become a necessity as well as a place to create new supporting devices. Sensor and computing technology has also penetrated into various aspects of human life. One of the important aspects is in terms of education or learning. This article will explore related to learning models that focus on the introduction of music, where music is something interesting and still has some problems in the learning process, including the introduction of symbolic notation, tone stairs, playing musical instruments, tones intonation and so on. Especially for the learning of early childhood preschool, a unique and interesting model is needed, in order to maintain consistency and final competence of learning. For this article, this article will present an alternative model of assistive devices in basic music learning for pre-school age children. A method of utilizing sensor technology and micro-control, packaged by utilizing TCS3200 and Arduino color sensors, is able to provide a unique solution for this music learning. The use of this type of color sensor, widely applied by previous researchers, such as its use of object sorting and enumeration [1] [2], detection of lamp lighting quality [3], laboratory fluid titration aids [4], there are even some researchers who also use the Visible Light Communication (VLC) method approach to conduct experiments to obtain recommendations for the use of lamps [5]. The learning model that fits the theme of this article is music learning, where music is able to provide concrete examples in the theme of Fourier analysis, signal processing and able to bridge the gap between education and contemporary research [6]. Game-based music learning with Game-Based Learning (GBL) methods is able to provide reinforcement of auditory learning outcomes with the aim of strengthening music intervals [7] [8]. Meanwhile, modeling a music concert conductor virtually [9], to an intelligent music player who is able to recommend based on the emotional connoisseur of his music [10], is able to provide an expansion of insights in the utilization of sensors, which coexist with educational interests in music learning into the future. Another modeling in support of basic music recognition, will be conveyed in this article, namely by the use of the TCS3200 color sensor, to detect colors according to the classification of tone sound categories and determine the appropriate sound exterior. A test simulation is done by making a color order, which is arranged in line in a song color band, so that it can be played well.

II. Previous work

Some of the research that has been done and focuses on music and sensors, will be the developer of insights this article, including an effort to model music notation in a 3D visual as one of the effective techniques for learning, where there are 3

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main attributes that build a music, namely tone, loudness and timbre, Designing a model of a setup tool also needs to be done to help the process of adaptive tuning in the microtonal arrangement of the music tool based on its tone key [11]. A system of recommendations for the selection of music, against the mood conditions of music lovers [12] [13], as well as modeling of modern virtual musical instruments [14] and traditional musical instruments involving artificial intelligence [15], artificial neural networks [16], representations of Fourier transformation [17], to the ability to create music with electronic instruments. The development of musical instruments has also been modified with the use of computer and sensor technology [18] [19] [20], as well as utilizing Internet of Things technology [21]. The use of Machine Learning [22], computing computer vision platforms [23] [24] , autorgressive modeling and amplitude factors [25], is able to serve fans and music lovers in interacting with machines through interactive karaoke systems [26].

III. Proposed System

A. Research Development

Meanwhile, one part of early childhood learning is the introduction of activities in color-based creativity, this is one of the rationale in this article. The modeling of eight unique colors, which is implemented for the basic tone of music in a unique way, is the use of color media from origami paper that is widely used and is expected to attract interest in play in children, especially at an early learning age. This is the target of the development of instrument models in this learning. The use of sensors that have specifications in detecting red, green and blue values based on Arduino micro control system, selected TCS3200 type sensor, which has been widely used to help system performance as in:

1) Classifying by color, on object [1] and fruit [2] in industrial-scale enterprises.
2) Detection of light color intensity, to predict the age of traffic lights [3].
3) Determination of the final process of titration of colored liquids in the laboratory [4].
4) Measurement of light intensity, to measure the comfort of the LED light user [5].

Based on this research, especially in the use and testing of the color sensor, the idea came up to utilize this sensor also to detect and calculate RGB values that appear, to be classified against 8 basic sounds of musical tones, namely do, re, mi fa, sol, la ti and do'.

B. Block diagram system

The system diagram as shown in figure 1, consists of an Arduino UNO microcontroller, with 5 digital I/O ports connected to the TCS3200 color sensor, 2 ports connected to the driver motor, and 1 port connected to the speaker. and apply the TCS3200 color sensor input to detect paper colors. The L298N is used to set the rotary speed of a DC motor at 60 revolution per minutes by utilizing the Pulse Width Modulation feature.

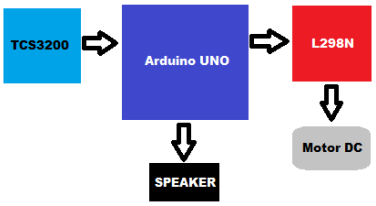

Fig. 1. Block diagram system

TABLE I
Input Control Direction of Actuator-B

IN3	IN4	Status
0	0	Stop
0	1	Backward
1	0	Forward
1	1	Stop

The forward-turning rotational conditioning status of the actuator connected to the L298N module, can be stated in table 1. Some of the basic formulas that are the circular movement calculation approach are,

$$Period\ T = 1/f \tag{1}$$

$$Circumference\ C = 2\pi r \tag{2}$$

$$Diameter\ d = 2r \tag{3}$$

$$Velocity\ v = C/T \tag{4}$$

The symbol f in the first formula is to express frequency, the symbol r in the second and third formulas expresses the radius of the circle, while the T symbol in the fourth formula is the time period. For settings C = 5 centimeters and T = 1 second, a value is required in the DC motor speed setting program code using the PWM feature, involving the setting program code based on high or low activation delay times.

C. Reference of RGB Color Data

The color category used as reference is as follows [27]:

Table 2 references several example colors as many as 16 colors, then we selected 8 basic colors, which will be connected with 8 basic tones as well as adjustments to the availability of colored paper materials in the researcher's environment. The colored paper based on origami paper, which has a wide variety of colors and is available in the market. Then the process of detecting, measuring and approaching color values to the material paper, is done using the TCS3200 sensor directly.

D. Development Stages

The main process of this color-based ladder selection system is the creation of color categories based on its color values in system training, and on this occasion there have been assigned 8 colors for 8 basic tone ladders, from do, re, mi, fa, sol, la, ti and do'. The color value acquisition data, which will be involved in the multilevel category function, is

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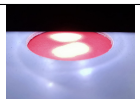
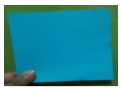
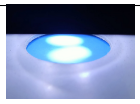
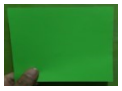
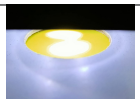
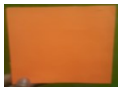
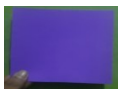
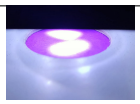
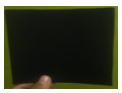
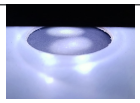
TABLE II
REFERENCE OF RGB COLOR CATEGORY

No	Color Value			Color Id
	R	*G*	*B*	
1	0	0	0	black
2	14	92	23	green
3	2	13	45	blue
4	169	25	77	hot pink
5	246	105	5	orange led
6	88	98	2	yellow
7	31	8	7	saddle brown
8	16	52	110	royal blue
9	30	85	36	forest green
10	30	18	41	indigo
11	113	151	51	lime green
12	92	33	91	purple
13	127	16	59	pink
14	159	19	24	dark red
15	106	109	122	grey
16	109	83	6	rod gold

obtained and can be observed in table 3, where detected and measurable color values are generated by the TCS3200 sensor, and measurement den detection is carried out 7 times for each selected paper color, according to the basic tone ladder classification.

TABLE III
RGB VALUE DETECTION RESULTS FOR TONE CLASSIFICATION

No.	Color Paper	Color Sensor	Color Value det	Tone
1				DO
2				RE
3				MI
4				FA
5				SOL
6				LA
7				TI
8				DO’

Some data results after color detection as summarized in table 3, obtaining eight color data to be selected into materials to be positioned on the color band, according to the ladder of basic musical rhythm notation tones. Paper will be affixed to a plastic tape according to the rhythm of the symbol notation on a song.

IV. RESULT

A. Implementation System and Algorithm

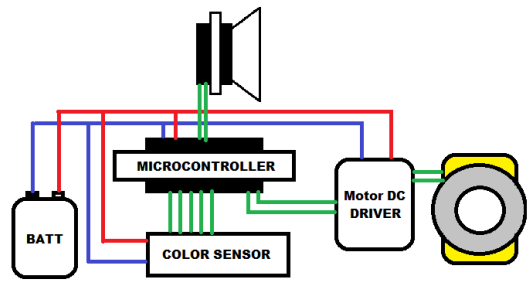


Fig. 2. The system diagram of color tone ladder

The system design diagram block is given as in figure 2, which includes the main parts of the arduino microcontroller, TCS3200 color sensor, DC motor driver, DC motor, speakers and battery resources. The prototype is designed to describe the design of the system, as in figure 3, which is visible front, side and top, where the color band of the tone ladder will be placed on a roller wheel connected to the DC motor, and this DC motor will be used to pull colored bands periodically at angular speeds at 1 lap. One spin is considered and tested to be able to represent a single tap of the tone.

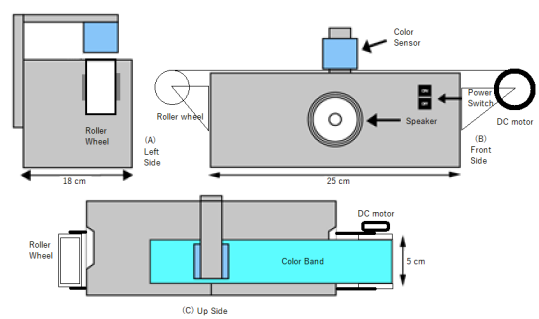


Fig. 3. The prototype of system model device

Figure 4 is the modeling and implementation of the circuit in system testing, which is operated and controlled automatically by the arduino main controller, based on the detection of the TCS3200 color sensor, and this circuit provides operational voltage power of 9 volts and current 1 Ampere. The motor settings are connected to the L298N module by utilizing the PWM feature to obtain a 60 rpm DC motor rotation setting duration.

When testing multiple samples of color bands according to figure 5, several determinants at once things that affect the color detection process on the ribbon, undergo significant changes, where the acquisition of data will change and be influenced by several things including:

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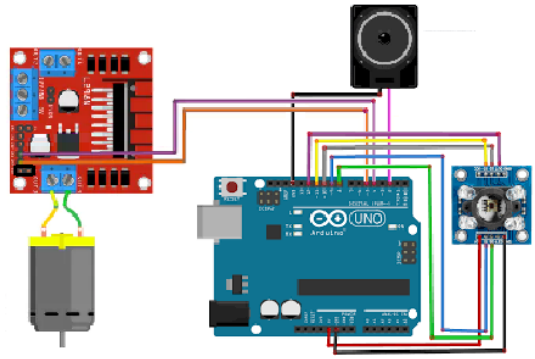

Fig. 4. The Circuit of color tone ladder.

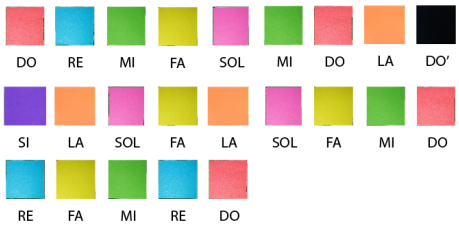

Fig. 5. The Color band testing performance system

1) Light intensity LED sensor
2) Stability of battery power source
3) Influence of ambient light intensity around the test
4) Color consistency on the test tape

Efforts to minimize misreading of color band data and maintain consistency, need to be done by approaching the average value of each color category and normalizing that value. The RGB color value classification can be observed in table 4.

TABLE IV
RGB VALUE CLASSIFICATION

No.	Color band	RGB Range Value					
		Rmin	Rmax	Gmin	Gmax	Bmin	Bmax
1	hot pink	66	93	122	149	86	111
2	blue	96	111	84	93	54	63
3	green	105	122	96	111	97	112
4	yellow	56	66	67	80	75	88
5	pink	45	45	60	66	38	44
6	orange led	38	41	74	89	66	77
7	purple	83	96	101	117	51	66
8	black	137	155	142	166	111	128

V. DISCUSSIONS

Figure 6 gives an overview of a piece of the national anthem ”Ibu Kita Kartini” with this simple combination of tones, used to represent the functional classification of the system. Then the number notation on the song will be made a line with the selected tone color, which is arranged on the color band in figure 7.

The color bands are arranged in a sequence that corresponds to the selected tone, rhythmically following the beat of the

Fig. 6. A Part of The Indonesian National Song ”Ibu Kita Kartini”s

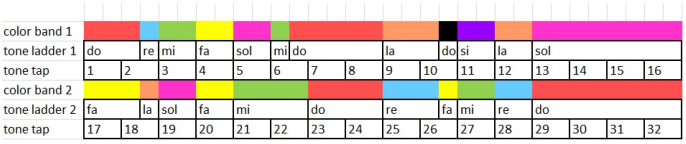

Fig. 7. Color Tone Ladder Device

tone with a period of every 1 second. The short length of the specific color band, determined by the number of tone beats, as well as the length of the selected tone ladder. Two color bands are intended to facilitate the reading and placement of appropriate colors only, but the realization is in the form of a row of colors one line of ribbon. The results of the implementation of this color-based tone ladder system have been tested and packaged in such a way as to obtain optimal conditions in color detection in a color band with the speed of tone taping in a period of one second. The device of color tone ladder can be observed in figure 8.

Fig. 8. Implementation of color tone ladder

VI. CONCLUSIONS

Modeling of musical instruments using ribbon model colored paper data, able to run well, through the process of paper color recognition and optimal placement of color sensors, able to dampen the reflection of light while minimizing the influence of interference during detection. The dimensions of the band and paper of 5 centimeters do not affect the stability of the DC motor rotation up to 60 rpm. Meanwhile, the length of the band and the long duration of time, will affect the stability of the dc motor’s rotary speed, to affect the notation of tone beats. The system’s temporary testing, conducted with a sample band length of 160 centimeters, was composed of 23 discoloration along 32 tone beats and was able to run well.

REFERENCES

[1] G. I. E. Panie and A. B. Mutiara, ”Development of Robotic Arm for Color Based Goods Sorter in Factory Using TCS3200 Sensor

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with a Web-Based Monitoring System," 2018 Third International Conference on Informatics and Computing (ICIC), 2018, pp. 1-6, doi: 10.1109/IAC.2018.8780461.

[2] D. Batra, H. Rewari and H. N, "Automated Tomato Sorting Machine," 2020 6th International Conference on Signal Processing and Communication (ICSC), 2020, pp. 206-210, doi: 10.1109/ICSC48311.2020.9182723.

[3] C. E. Ngene and T. Shongwe, "A Sensor for Monitoring the Lifespan of Color-LEDs in Traffic Lights," 2019 International Conference on Advances in Big , Computing and Data Communication Systems (icABCD), 2019, pp. 1-5, doi: 10.1109/ICABCD.2019.8851019.

[4] Yao Li and Lei Zhang, "An automatic determination method based on color sensor at the end point of the titration," 2011 Second International Conference on Mechanic Automation and Control Engineering, 2011, pp. 3836-3838, doi: 10.1109/MACE.2011.5987836.

[5] C. E. Ngene, P. Thakur and G. Singh, "Visible Light Communication-LEDs Illuminations for Smart Homes," 2020 International Conference on Artificial Intelligence, Big Data, Computing and Data Communication Systems (icABCD), 2020, pp. 1-5, doi: 10.1109/icABCD49160.2020.9183871.

[6] M. Mueller, B. McFee and K. Kinnaird, "Interactive Learning of Signal Processing Through Music: Making Fourier Analysis Concrete for Students," in IEEE Signal Processing Magazine, vol. 38, no. 3, pp. 73-84, May 2021, doi: 10.1109/MSP.2021.3052181.

[7] M. I. Rizqyawan and G. Hermawan, "Adventure game as learning media for introducing music interval and ear training to kids," 2015 International Conference on Automation, Cognitive Science, Optics, Micro Electro-Mechanical System, and Information Technology (ICACOMIT), 2015, pp. 172-175, doi: 10.1109/ICACOMIT.2015.7440200.

[8] P. Raziūnaitė, A. Miliūnaitė, R. Maskeliūnas, R. Damaševičius, T. Sidekerskienė and B. Narkevičienė, "Designing an educational music game for digital game based learning: A Lithuanian case study," 2018 41st International Convention on Information and Communication Technology, Electronics and Microelectronics (MIPRO), 2018, pp. 0800-0805, doi: 10.23919/MIPRO.2018.8400148.

[9] A. Nijholt, D. Reidsma, R. Ebbers and M. t. Maat, "The Virtual Conductor: Learning and Teaching about Music, Performing, and Conducting," 2008 Eighth IEEE International Conference on Advanced Learning Technologies, 2008, pp. 897-899, doi: 10.1109/ICALT.2008.43.

[10] S. Gilda, H. Zafar, C. Soni and K. Waghurdekar, "Smart music player integrating facial emotion recognition and music mood recommendation," 2017 International Conference on Wireless Communications, Signal Processing and Networking (WiSPNET), 2017, pp. 154-158, doi: 10.1109/WiSPNET.2017.8299738

[11] K. Stange, C. Wick and H. Hinrichsen, "Playing Music in Just Intonation: A Dynamically Adaptive Tuning Scheme," in Computer Music Journal, vol. 42, no. 3, pp. 47-62, Oct. 2018, doi: 10.1162/comj_a_00478.method.

[12] I. A. Pegoraro Santana et al., "Music4All: A New Music Database and Its Applications," 2020 International Conference on Systems, Signals and Image Processing (IWSSIP), 2020, pp. 399-404, doi: 10.1109/IWSSIP48289.2020.9145170.

[13] K. M. Ibrahim, J. Royo-Letelier, E. V. Epure, G. Peeters and G. Richard, "Audio-Based Auto-Tagging With Contextual Tags for Music," ICASSP 2020 - 2020 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), 2020, pp. 16-20, doi: 10.1109/ICASSP40776.2020.9054352.

[14] J. A. Paradiso, "Electronic music: new ways to play," in IEEE Spectrum, vol. 34, no. 12, pp. 18-30, Dec. 1997, doi: 10.1109/6.642965.

[15] F. Yang, "Artificial Intelligence in Music Education," 2020 International Conference on Robots & Intelligent System (ICRIS), 2020, pp. 483-484, doi: 10.1109/ICRIS52159.2020.00124.

[16] A. A. Mehta and M. S. Bhatt, "Optical music notes recognition for printed piano music score sheet," 2015 International Conference on Computer Communication and Informatics (ICCCI), 2015, pp. 1-6, doi: 10.1109/ICCCI.2015.7218061.

[17] G. Costantini, M. Todisco and R. Perfetti, "A novel sensor interface for detecting musical notes of percussive pitched instruments," 2009 3rd International Workshop on Advances in sensors and Interfaces, 2009, pp. 121-126, doi: 10.1109/IWASI.2009.5184781.

[18] A. Kapur, E. L. Yang, A. R. Tindale and P. F. Driessen, "Wearable sensors for real-time musical signal processing," PACRIM. 2005 IEEE Pacific Rim Conference on Communications, Computers and signal Processing, 2005., 2005, pp. 424-427, doi: 10.1109/PACRIM.2005.1517316.

[19] G. Saggio, F. Giannini, M. Todisco and G. Costantini, "A data glove based sensor interface to expressively control musical processes," 2011 4th IEEE International Workshop on Advances in Sensors and Interfaces (IWASI), 2011, pp. 192-195, doi: 10.1109/IWASI.2011.6004715.

[20] J. -H. Jun, Sunardi, L. Wang, J. W. Matthys, S. Lui and Y. Gu, "Demo abstract: Sensor-enabled Yo-yos as new musical instruments," 2012 ACM/IEEE 11th International Conference on Information Processing in Sensor Networks (IPSN), 2012, pp. 135-136, doi: 10.1109/IPSN.2012.6920933.

[21] N. Pan, "Research on Music Wireless Control Based on Motion Tracking Sensor and Internet of Things," in IEEE Access, vol. 9, pp. 48803-48810, 2021, doi: 10.1109/ACCESS.2021.3064565.

[22] H. T. Panduranga and C. Mani, "Sensors for virtual musical environment: A short survey," 2017 International Conference on Electrical, Electronics, Communication, Computer, and Optimization Techniques (ICEECCOT), 2017, pp. 148-157, doi: 10.1109/ICEECCOT.2017.8284656.

[23] G. Odowichuk, S. Trail, P. Driessen, W. Nie and W. Page, "Sensor fusion: Towards a fully expressive 3D music control interface," Proceedings of 2011 IEEE Pacific Rim Conference on Communications, Computers and Signal Processing, 2011, pp. 836-841, doi: 10.1109/PACRIM.2011.6033003.

[24] N. Burks, L. Smith and J. Saquer, "A Virtual Xylophone for Music Education," 2016 IEEE International Symposium on Multimedia (ISM), 2016, pp. 409-410, doi: 10.1109/ISM.2016.0094.

[25] R. Blouet, G. Rapaport, I. Cohen and C. Fevotte, "Evaluation of several strategies for single sensor speech/music separation," 2008 IEEE International Conference on Acoustics, Speech and Signal Processing, 2008, pp. 37-40, doi: 10.1109/ICASSP.2008.4517540.

[26] K. Minamoto and K. Yoshino, "Development of Hand-Held Digital Effector Using Touch Sensor," 2017 5th Intl Conf on Applied Computing and Information Technology/4th Intl Conf on Computational Science/Intelligence and Applied Informatics/2nd Intl Conf on Big Data, Cloud Computing, Data Science (ACIT-CSII-BCD), 2017, pp. 65-70, doi: 10.1109/ACIT-CSII-BCD.2017.53.

[27] D. Ratnawati and V. Vivianti, "Education Game Kit "Princess Stick and Color Sword" For Preschooler by Using TCS3200 Color Censor and Arduino Nano," 2019 International Journal of Applied Business and Information Systems, 2019, pp. 9–13, doi:10.31763/ijabis.v3i1.158

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